

Template Title

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Contents

$$\left[\frac{d^2}{dr^2} + \frac{\frac{1}{4} - (2j+1)^2}{r^2} + 4 \left(j + \frac{1}{\lambda} \right) ea - e^2 r^2 a^2 + \frac{e}{\lambda} r \frac{da}{dr} \right] = m^2 f, \quad (1)$$

For $r > R$ let $f = r^{\frac{1}{2}} g$ so that (2) becomes

$$g'' + \frac{1}{r} g' - \left(m^2 + \frac{(2j+1 - e\nu)^2 + 2 \left(1 - \frac{1}{\lambda} \right) e\nu}{r^2} \right) g = 0. \quad (2)$$

This has the decaying solution $K_\alpha(mr)$ which is a modified Bessel function with $\alpha = \left[(2j+1 - e\nu)^2 + 2 \left(1 - \frac{1}{\lambda} \right) e\nu \right]^{\frac{1}{2}}$. The eigenvalue λ is fixed by the boundary condition at $r = R$:

$$\frac{Rf'(R)}{f(R)} = \frac{1}{2} + \frac{RK'_\alpha(mR)}{K_\alpha(mR)} \quad (3)$$

The analysis is simplified by setting $mR \ll 1$ and making the small mass expansions $F = Bf_0 (1 + m^2 f_2 + \mathcal{O}(m^4 f_4))$, $\lambda = 1 - m^2 \delta_2 + \mathcal{O}(m^4 \delta_4)$, B is a normalization constant. Here f_0 is the solution of (2) when $m = 0$ and $\lambda = 1$ and $0 \leq r \leq R$ where $f_0 = r^{2j+\frac{3}{2}} e^{-e \int_0^r ds sa(s)}$. The two sides of (3) can be matched in powers of m to obtain for $e\nu \gg 1$

$$\lambda = 1 - \frac{m^2 R^2}{2e\nu} \left(1 + \mathcal{O} \left(\frac{1}{e\nu} \right) \right), \quad (4)$$

following Appendix E of [Fry2015Electroweak]. Such an eigenvalue approaching 1 from below is the signature of a QZM as referred to in Sec. 1 above.

We conclude that for this mode its contribution to $\ln \det_3$ in (??) is

$$\ln \det_3 = -\frac{e^2 \nu^2}{2} \ln \left(\frac{e\nu}{m^2 R^2} \right) \left(1 + \mathcal{O} \left(\frac{1}{e\nu} \right) \right) + \sum_{n=2}^{\infty} [\ln(1 - \lambda_n^2) + \lambda_n^2] \quad (5)$$